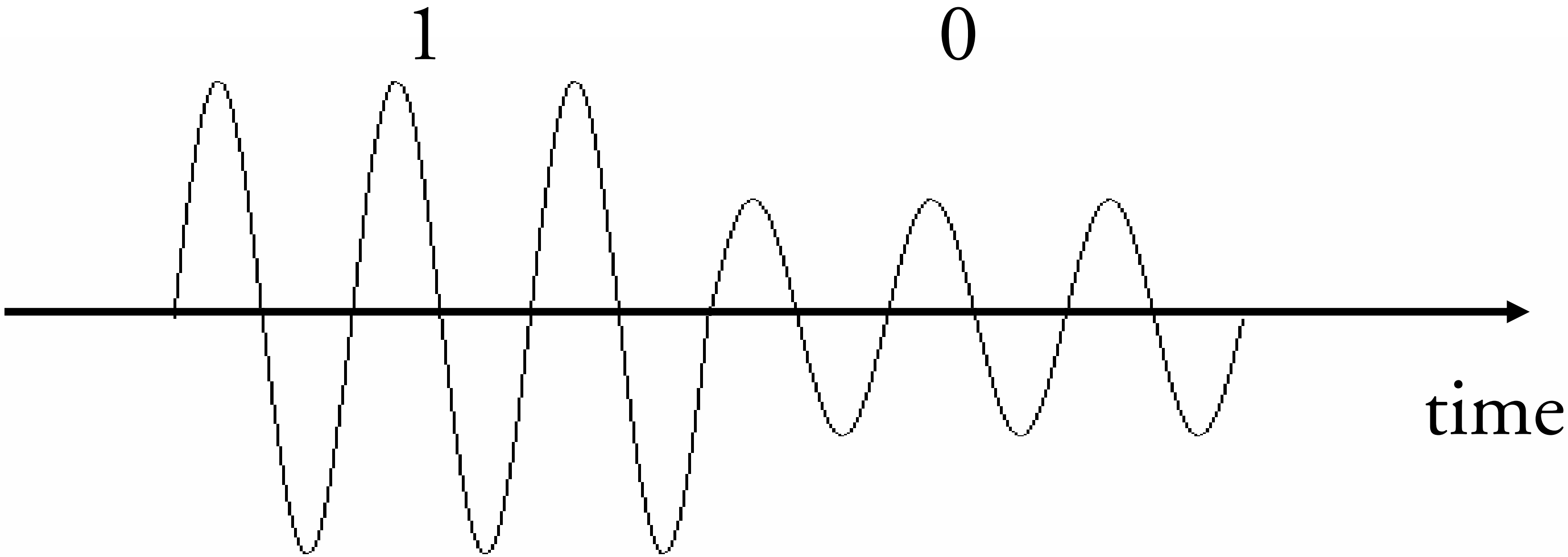
Midterm exam 061113

Solutions

1: Name the three steps in PCM

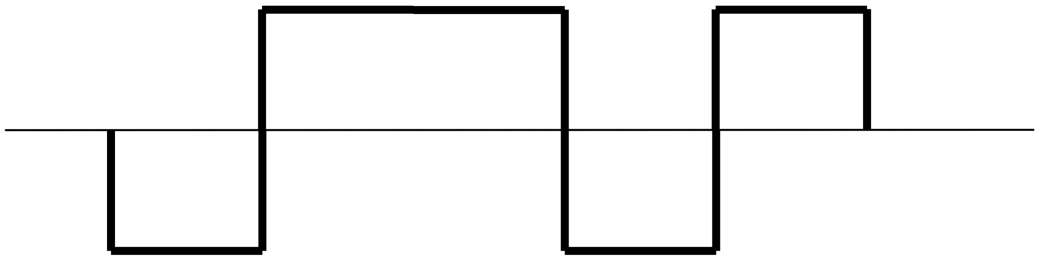
Sampling, Quantization, Encoding

2: Show how 0s and 1s can be modulated with ASK!



3. What is the main problem with the NRZ coding scheme?

Synchronization!



4.What is the objective of the MAC-protocol in the IEEE 802.x standards?

The MAC-protocol determines the procedures when a station needs *access to the transmission medium*.

Examples of MAC-protocols are Token passing and CSMA/CD.

5: ALOHA is a so called random access method. What does this mean?

It means that each station individually determines when it should send data on the link. No station is controlling the other stations.

6. What transmission medium is used in IEEE 802.11?

Air!

7. What is cross-talk (överhörning)?

It is the transmisson impairment caused by interference between two wires that are close to each other.

8: Explain is the main difference between Go-back-N and Selective Repeat ARQ.

In Go-back-N, the receiver ACKs all frames correctly received *in the right order*. When an error occurs, the *sender retransmits all frames* not ACKed.

In Selective Repeat, the receiver *sends NAKs when it detects a missing frame*. This means that the sender can minimize its retransmissions.

9: Calculate a 2-bit CRC for the bit sequence 1101. The generator polynomial is x^2+1.

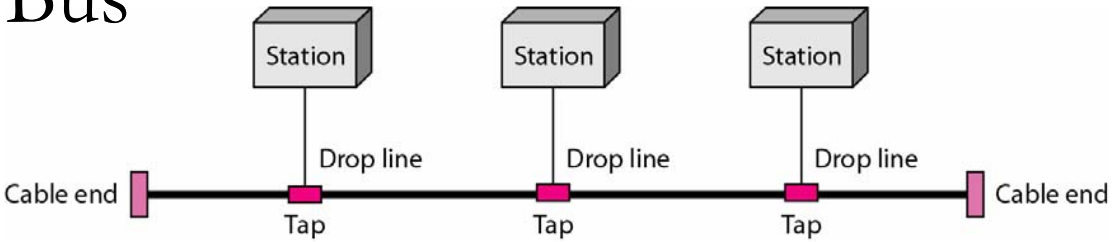
1101 corresponds to the polynomial x^3+x^2+1.

The CRC is the remainder in the division: $x^5+x^4+x^2$ / x^2+1.

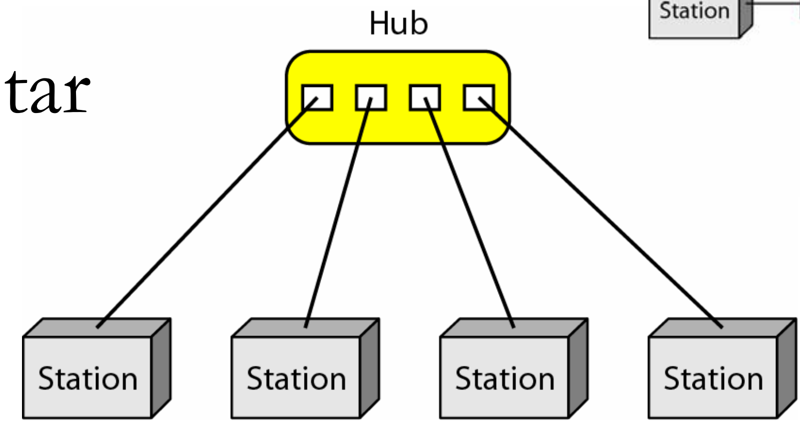
The CRC becomes 10 (x).

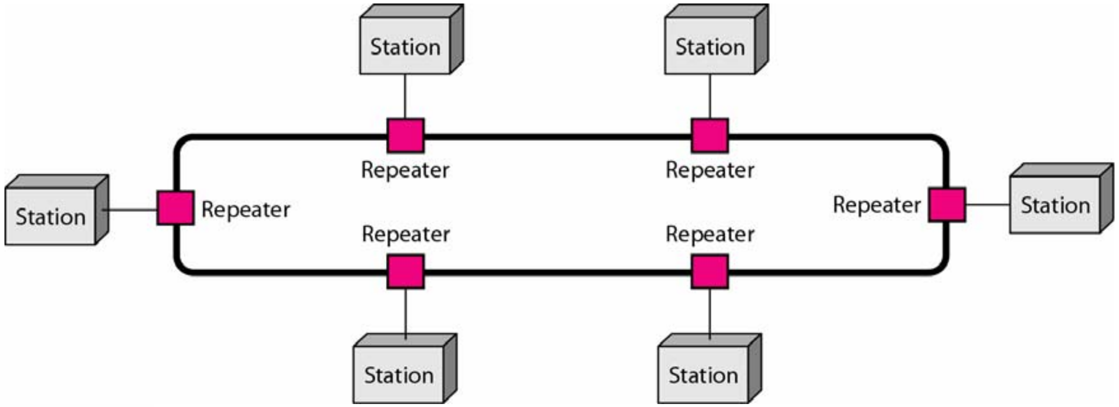
10: Show and name two network topologies for wired LANs.

Bus



Star





Ring